

OpenMission

Preliminary Mission Analysis Tool for
SSO Earth Observation Constellations

Technical Documentation

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github.com/ProfPyg/OpenMission

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1 Overview

OpenMission is a browser-based tool for preliminary mission analysis of Sun-Synchronous Orbit (SSO) constellations for Earth Observation (EO) applications. It is organised into three modules, each accessible as a separate page in the Streamlit application:

1. **SSO Coverage Analysis** — computes coverage and revisit metrics over a user-defined Region of Interest (ROI) and access windows for one or more ground stations, using the Skyfield/SGP4 propagator with WGS84 ellipsoid geometry.
2. **Orbital Lifetime Estimator** — estimates the time a satellite remains in orbit before atmospheric drag causes decay below 80 km, using NRLMSISE-00 with ECSS-E-ST-10-04C solar activity scenarios. Supports natural decay, two-burn circularization (Hohmann transfer), and continuous thrust.
3. **Pass Prediction** — predicts ground station contact windows for an existing satellite defined by a TLE, using Skyfield/SGP4. Includes TLE age warnings based on expected position accuracy.

The tool is deployed as a Streamlit web application. All orbital computations are performed server-side in Python. Geographic border data is hardcoded — no network request is made at runtime.

OpenMission is intended for early trade-space exploration. Results should be verified against higher-fidelity tools (STK, MATLAB Aerospace Toolbox, ESA DRAMA) before use in final design.

2 SGP4 Propagator

The SGP4 propagator is shared by the SSO Coverage Analysis and Pass Prediction modules. Satellites are initialised from Keplerian elements (SSO Coverage) or TLE data (Pass Prediction) using the Skyfield library [9] with the `sgp4` backend [10].

2.1 Satellite Initialisation from Keplerian Elements

For the SSO Coverage module, satellites are built from clean Keplerian elements using `sgp4.api.Satrec.sgp4init` with $B^* = 0$ (no atmospheric drag):

```
satrec = Satrec()
satrec.sgp4init(WGS84, 'i', satnum, epoch_days,
                0.0, 0.0, 0.0, # ndot, nddot, bstar
                ecc, argp_rad, inc_rad,
                M0_rad, n_rad_per_min, RAAN_rad)
sat = EarthSatellite.from_satrec(satrec, ts)
```

2.2 Satellite Initialisation from TLE

For the Pass Prediction module, satellites are built directly from TLE strings:

```
satellite = EarthSatellite(line1, line2, name, ts)
```

2.3 Mean Motion

Derived from the semi-major axis [10, Eq. 1]:

$$n = 60 \cdot \sqrt{\frac{\mu}{a^3}} \quad [\text{rad/min}] \quad (1)$$

2.4 Epoch

The SGP4 epoch is expressed as days since 1949 December 31 00:00 UT [10, §3.1]:

$$t_{\text{epoch}} = \frac{(t_{\text{start}} - 1949-12-31T00:00:00)_{\text{total seconds}}}{86400} \quad (2)$$

2.5 Ground Point Geometry

All ground stations and ROI points are constructed on the WGS84 oblate spheroid via `skyfield.api.wgs84.latlon` which correctly models Earth’s flattening ($f = 1/298.257$). Topocentric elevation angles are computed via `topocentric.altaz()`, which includes full IAU-76/FK5 Earth rotation (precession and nutation).

3 SSO Coverage Analysis

3.1 Inputs

3.1.1 Constellation

Parameter	Type	Description
Orbital planes P	User	Number of orbital planes
Satellites per plane S	User	Satellites per plane
Total satellites N	Derived	$N = P \times S$

Satellites are equally spaced within each plane and planes are equally spaced in RAAN:

$$M_0(s) = s \cdot \frac{360}{S} \quad [\text{deg}] \quad (3)$$

$$\Omega(p) = \Omega_0 + p \cdot \frac{360}{P} \quad [\text{deg}] \quad (4)$$

3.1.2 Classical Orbital Elements

Element	Symbol	Input	Value / Formula
Semi-major axis	a	Derived	$a = R_e + h$, $R_e = 6378.137$ km
Eccentricity	e	Locked	0.001
Inclination	i	Auto/Manual	Section 3.1.3
RAAN (plane 1)	Ω_0	User	Live-linked to LTAN
LTAN	LTAN	User	Live-linked to Ω_0
Arg. of perigee	ω	Locked	0
Mean anomaly	M_0	Locked	0

RAAN and LTAN are live-linked via the Sun’s right ascension α_s [1, §5.1, Eq. 5-4 to 5-7]:

$$\Omega_0 = \text{mod}(\alpha_s + 15(\text{LTAN} - 12), 360) \quad (5)$$

$$\text{LTAN} = \text{mod}\left(12 + \frac{\Omega_0 - \alpha_s}{15}, 24\right) \quad (6)$$

3.1.3 SSO Inclination

In SSO Auto mode [1, §11.4]:

$$\cos i = -\frac{\dot{\Omega}_{\odot}}{\frac{3}{2} n J_2 \left(\frac{R_e}{a}\right)^2}, \quad \dot{\Omega}_{\odot} \approx 0.9856 \text{ deg/day} \quad (7)$$

3.1.4 Simulation Window

Parameter	Type	Description
Start date & time	User	UTC
Duration	User	1–365 days
Sample step	User	30 s (fine), 60 s (standard), 120 s (fast)
Repeat ground track	Derived	Exact repeat cycle (Section 3.3)

Note: If the exact repeat cycle exceeds 30 days, OpenMission finds the closest integer number of days within 30 days that minimises the residual non-integer orbital count.

3.1.5 Payload Constraint

A ground point is accessible only if the off-nadir angle satisfies [3, §9.1]:

$$\theta = \arccos\left(\hat{r}_{\text{sat}} \cdot \frac{-\Delta\mathbf{r}}{|\Delta\mathbf{r}|}\right) \leq \theta_{\max} \quad \Leftrightarrow \quad \text{el} \geq 90 - \theta_{\max} \quad (8)$$

Ground stations use the elevation constraint only.

3.1.6 Ground Stations

Each ground station is defined by name, latitude, longitude, altitude, and minimum elevation angle. Two pre-loaded defaults:

Name	Lat [°N]	Lon [°E]	Alt [m]	Min El [°]
Helmos Observatory, NOA	37.9856	22.1983	2340	30
UoA Ground Station, Psachna	38.5697	23.6486	80	5

3.1.7 Region of Interest

The user selects a country from 61 supported countries (all European states plus selected Mediterranean countries). OpenMission generates a uniform grid over the bounding box and filters points inside the border using the ray-casting algorithm [12]. A custom shapefile (.shp + .dbf + .shx) can be uploaded for higher-resolution borders. Country borders are from Natural Earth 1:110m [13].

3.2 Access and Revisit Computation

3.2.1 Elevation Angle

$$\text{el} = \arcsin\left(\frac{(\mathbf{r}_{\text{sat}} - \mathbf{r}_{\text{gs}}) \cdot \hat{n}_{\text{gs}}}{|\mathbf{r}_{\text{sat}} - \mathbf{r}_{\text{gs}}|}\right) \quad (9)$$

[1, §4.4, Eq. 4-4]

3.2.2 Window Detection and Merging

Boolean visibility arrays are computed across all timesteps and satellites. Windows are merged using the same algorithm as `VLEOExample.m`:

```
currentStart, currentEnd = windows[0]
for start, end in windows[1:]:
    if start <= currentEnd:
        currentEnd = max(currentEnd, end)
    else:
```

```
merged.append((currentStart, currentEnd))
currentStart, currentEnd = start, end
merged.append((currentStart, currentEnd))
```

3.2.3 Revisit Metrics

$$g_i = t_{\text{start},i+1} - t_{\text{end},i} \quad (10)$$

$$\bar{g} = \text{mean}(g_i), \quad g_{\text{max}} = \text{max}(g_i) \quad (11)$$

$$\text{Access ratio} = \frac{\sum \Delta t_i}{T_{\text{sim}}} \times 100\% \quad (12)$$

3.3 Repeat Ground Track Cycle

A repeating ground track requires [1, §11.5]:

$$\frac{1440}{T[\text{min}]} = \frac{p}{q}, \quad \text{gcd}(p, q) = 1 \quad (13)$$

The repeat cycle is q days. Simulating for less than one cycle may underestimate revisit times.

3.4 Time-Step Quantisation Error

Step Δt	Edge uncertainty	Mean revisit error
30 s	± 15 s	~ 1 min
60 s	± 30 s	~ 2 min
120 s	± 60 s	~ 4 min

3.5 Validation

3.5.1 Propagator Function Verification

At epoch 2026-03-02 08:23:00 UTC:

Function	MATLAB	OpenMission	Status
sso_inclination(500 km)	97.4016°	97.4016°	✓
ltan2raan(epoch, 10.5 h)	320.6841°	320.6841°	✓
raan2ltan(epoch, 320.6841°)	10.5000 h	10.5000 h	✓

3.5.2 Ground Station Validation vs MATLAB

Configuration: 3×3, 500 km SSO, LTAN=10.5 h, epoch 2026-03-02, 9-day simulation, 60 s step.

Metric	MATLAB	OpenMission	Diff
Helmos access ratio	2.01%	1.58%	0.43%
Helmos mean revisit	1h 49m 37s	1h 46m 49s	~ 3 min ✓
Helmos max revisit	4h 36m 30s	4h 37m 00s	~ 30 s ✓
Psachna access ratio	16.75%	15.46%	1.29%
Psachna mean revisit	33m 48s	34m 05s	~ 17 s ✓
Psachna max revisit	2h 28m 30s	2h 29m 00s	~ 30 s ✓

Access ratio differences (~ 0.4 – 1.3%) are attributed to the $\sim 3^\circ$ RAAN offset between MATLAB and OpenMission (different solar position models).

3.5.3 ROI Single-Point Validation vs MATLAB and STK

Athens (37.9856°N, 23.7275°E), off-nadir ≤ 45 , 9-day simulation.

Metric	MATLAB	STK	OpenMission	Diff
Mean revisit	0h 23m 58s	0h 23m 58s	0h 24m 17s	~ 19 s ✓
Max revisit	1h 55m 00s	1h 55m 00s	1h 56m 00s	~ 1 min ✓
Access ratio	26.43%	26.43%	24.95%	1.48%

All three tools consistent within time-step quantisation error.

4 Orbital Lifetime Estimator

4.1 Inputs

Parameter	Type	Description
Dry mass	User	Satellite mass without propellant [kg]
Cross-sectional area	User	A [m ²]
Drag coefficient C_D	User	Default 2.2
Altitude	User	Initial circular orbit [km], max 500 km
Solar activity	User	Solar Min / Mean / Max / All three
Propellant mass	User	Available for deorbit [kg] (if propulsion enabled)
Thrust T	User	[mN] (if propulsion enabled)
Specific impulse I_{sp}	User	[s] (if propulsion enabled)
Deorbit strategy	User	Natural / Two-burn circularization / Continuous thrust
Deorbit threshold	Fixed	80 km (DRAMA OSCAR convention)

Derived: ballistic coefficient $\beta = m/(C_D A)$ [kg/m²].

4.2 Atmosphere Model — NRLMSISE-00

Atmospheric density is computed using NRLMSISE-00 [8], sampled at $\phi = 0, \lambda = 0$ (equatorial approximation) using the current date as epoch. Density is returned in g/cm³ and converted to SI:

$$\rho \text{ [kg/m}^3\text{]} = \text{result}[5] \times 1000 \quad (14)$$

4.3 Solar Activity Scenarios

Following ECSS-E-ST-10-04C Rev.1 [14], Tables G-1 to G-3:

Scenario	F10.7	F10.7a	Ap
Solar Min	65	65	0
Solar Mean	140	140	15
Solar Max	250	250	45

4.4 Drag Equation and Numerical Integration

Semi-major axis decay rate [1, 5]:

$$\frac{da}{dt} = -C_D \frac{A}{m} \rho \sqrt{\mu a} \quad [\text{m/s}] \quad (15)$$

Integrated using 4th-order Runge-Kutta with adaptive step:

$$\begin{aligned}\Delta t &= T_{\text{orb}} \quad (h \geq 200 \text{ km}) \\ \Delta t &= 0.1 T_{\text{orb}} \quad (h < 200 \text{ km})\end{aligned}$$

Decay threshold: fixed at 80 km (ESA DRAMA OSCAR convention).

4.5 Propulsion Model

4.5.1 Two-Burn Circularization (Hohmann Transfer)

Two retrograde impulses: the first lowers perigee to target altitude, the second circularizes the orbit at target altitude. This matches the ESA DRAMA OSCAR re-orbit strategy.

$$\Delta V_1 = v_{\text{circ}}(r_1) - v_{\text{ap}}(r_1, a_{\text{tr}}) \quad (\text{perigee drop}) \quad (16)$$

$$\Delta V_2 = v_{\text{circ}}(r_2) - v_{\text{pe}}(r_2, a_{\text{tr}}) \quad (\text{circularization}) \quad (17)$$

$$\Delta V_{\text{total}} = \Delta V_1 + \Delta V_2 \quad [1, 6.3] \quad (18)$$

where $a_{\text{tr}} = (r_1 + r_2)/2$ is the Hohmann transfer SMA, $r_1 = R_e + h_0$ and $r_2 = R_e + h_{\text{target}}$.

Propellant computed by applying Tsiolkovsky twice in series [6]:

$$m_1 = m_{\text{wet}} e^{-\Delta V_1 / (I_{\text{sp}} g_0)} \quad (19)$$

$$m_2 = m_1 e^{-\Delta V_2 / (I_{\text{sp}} g_0)} \quad (20)$$

$$m_{\text{prop}} = m_{\text{wet}} - m_2 \quad (21)$$

Post-burn orbit: circular at h_{target} . Integration starts from r_2 (circular orbit at target altitude).

Validation note: For 430 km $\rightarrow$ 200 km, DRAMA OSCAR reports $\Delta V = 132.6$ m/s (two-burn), while OpenMission two-burn gives ≈ 132 m/s. ✓

4.5.2 Continuous Thrust

$$\frac{da}{dt} = \left. \frac{da}{dt} \right|_{\text{drag}} - \frac{2 a T}{m v_{\text{circ}}} \quad (22)$$

$$\frac{dm}{dt} = -\frac{T}{I_{\text{sp}} g_0} \quad [6] \quad (23)$$

4.6 5-Year Compliance Check

Lifetime	Status
≤ 5 years	Compliant
> 5 years	Non-compliant

References: ESA ESSB-ST-U-007 Issue 1 (2023) [15]; FCC Report and Order (2022) [17]; IADC-02-01 Rev.4 (2024) [16].

4.7 Validation vs ESA DRAMA OSCAR

Natural decay, Solar Mean (F10.7 = 140, Ap = 15), $m = 5.5$ kg, $A = 0.03 \text{ m}^2$, $C_D = 2.2$, epoch 2026-06-04, deorbit threshold 80 km.

Alt.	DRAMA	OSCAR	OpenMission	Diff.	Note
(km)	Days	Years	Days	Years	
200	1	<0.01	2	<0.01	— abs. diff < 1 day
250	14	0.038	11	0.03	— abs. diff < 1 weeks
300	52	0.14	41	0.11	— abs. diff < 2 weeks
350	139	0.38	130	0.36	6.5% ✓
400	339	0.93	366	1.00	8.0% ✓
450	989	2.71	963	2.64	2.6% ✓
500	1988	5.44	2405	6.58	21% boundary of model validity

Table 1: Orbital lifetime validation against ESA DRAMA OSCAR. Natural decay, Solar Mean, $m = 5.5$ kg, $A = 0.03$ m², $C_D = 2.2$, epoch 2026-06-04, deorbit threshold 80 km.

At altitudes of 350–450 km, agreement is within 10% (2.6–8.0%). At low altitudes (≤ 300 km) percentage differences appear large but are physically insignificant: the absolute error is less than two weeks, and both tools agree the satellite re-enters within days to weeks — a distinction irrelevant for mission planning. At 500 km the difference is 21% (417 days); this altitude marks the upper boundary of reliable NRLMSISE-00 accuracy for this tool, and results there should be treated as indicative. The tool enforces a 500 km altitude ceiling and displays a warning at that limit. The systematic differences are attributed to the equatorial atmosphere approximation ($\phi = 0$) and the absence of geopotential perturbations (DRAMA uses GEM-T1 6×6, OpenMission uses drag-only integration).

Propulsion validation (two-burn Hohmann circularization, 500 km $\rightarrow$ 200 km, $T = 50$ mN, $I_{sp} = 1200$ s, $m_{prop} = 0.5$ kg, $m = 5.5$ kg, $A = 0.03$ m², $C_D = 2.2$, epoch 2026-06-04):

Metric	DRAMA	OpenMission	Diff	Status
ΔV_{total}	171.63 m/s	171.63 m/s	0%	✓
Propellant consumed	0.08 kg	0.0869 kg	<0.01 kg	✓
Post-burn deorbit date	2026-06-07	2026-06-07	0 days	✓

5 Pass Prediction

5.1 Inputs

Parameter	Type	Description
TLE file	Upload	Standard 2-line or 3-line format (.txt)
Ground location	User	Preset station or custom lat/lon
Min elevation	User	Minimum elevation angle [°], default 5°
Days ahead	User	1, 2, 3, 7, or 14 days from TLE epoch

The simulation starts at the TLE epoch (most accurate point). The user selects how many days ahead to predict passes.

5.2 Computation

The satellite is initialised directly from TLE strings via Skyfield (Section 2). The same elevation check and window detection as the SSO Coverage module are used. A 10-second sample step is used (single satellite — negligible computation cost) for accurate peak elevation capture.

5.3 TLE Accuracy and Age Warnings

TLE position accuracy degrades over time [10, 11]. Timing errors are derived from position errors using LEO orbital velocity ~ 7.6 km/s:

TLE Age	Position Error	Timing Error	Warning
< 3 days	<~1 km	<~10 s	Good
3–7 days	~1–5 km	<~1 min	Acceptable
7–14 days	~5–20 km	<~5 min	Caution
> 14 days	>20 km	>5 min	Update recommended

Accuracy depends on satellite altitude and atmospheric drag environment. High-altitude or low-drag satellites may remain accurate beyond these guidelines.

5.4 Validation vs Heavens-Above

ISS (ZARYA, NORAD 25544) TLE downloaded from Space-Track and fed into OpenMission. Heavens-Above used its own independently sourced TLE for the same satellite. Ground location: Helmos Observatory (37.9856°N, 22.1983°E, 2340 m). Minimum elevation: 10°. Period: 2026-06-02 to 2026-06-08.

Metric	Value
Matched passes	32 / 45
Mean timing difference	<0.1 min (~ 6 s)
Maximum timing difference	0.1 min (~ 6 s)
Mean max elevation difference	<1°

Selected passes:

#	HA Start (UTC)	OM Start (UTC)	HA El	OM El	Diff
1	08:40:25	08:40:28	42°	41.6°	3 s
2	10:17:44	10:17:48	23°	22.9°	4 s
3	15:10:57	15:10:58	31°	30.6°	1 s
6	09:29:16	09:29:18	34°	34.3°	2 s
11	08:41:00	08:41:08	58°	58.3°	8 s
14	15:11:30	15:11:38	88°	86.8°	8 s
15	07:52:58	07:52:58	77°	76.8°	0 s
29	07:05:34	07:05:38	61°	60.9°	4 s
32	13:36:03	13:36:08	84°	82.6°	5 s

Table 2: ISS pass prediction: Heavens-Above vs OpenMission
(2026-06-02 to 2026-06-08)

The 13 unmatched passes fall outside the 7-day simulation window — not a discrepancy.

Note on ERMIS-3 validation: A further validation using ERMIS-3 TLE files is planned, leveraging direct access through participation in the ERMIS mission team. This will provide validation against a non-maneuvering SSO satellite more representative of the target use case.

6 Limitations

1. **Propagator.** SGP4 with $B^* = 0$ does not model atmospheric drag, solar radiation pressure, or lunar/solar gravity in the access module.
2. **Polygon resolution.** Natural Earth 1:110m borders may misclassify points near coastlines. Upload a custom shapefile for higher resolution.
3. **Atmosphere model.** NRLMSISE-00 with fixed solar indices. Real density varies by $\pm 10\%$; future solar activity is unpredictable.
4. **Circular orbit.** Eccentricity locked at 0.001. Elliptical orbits not supported.
5. **No swath width.** ROI access is point-target only. Area coverage requires swath width, not currently modelled.
6. **Single ROI country.** One country per analysis run.
7. **Equatorial atmosphere approximation.** Lifetime Estimator samples NRLMSISE-00 at $\phi = 0$, overestimating lifetime for high-inclination (SSO) orbits above ~ 400 km.
8. **Altitude ceiling.** Lifetime Estimator is limited to 500 km. Results at this boundary carry a $\sim 21\%$ deviation from DRAMA OSCAR and should be treated as indicative only.

7 License and Citation

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<https://github.com/ProfPyg/OpenMission>

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A User Guide

This appendix is a practical, screen-by-screen walkthrough of the application. It complements the technical sections above: where those describe *what* each module computes, this guide describes *how* to drive it. The values shown in the figures form a single worked example — a 3×3 Walker constellation at 500 km in a Sun-synchronous orbit — so the screenshots can be read in sequence as one continuous configuration.

OpenMission is organised as three pages, selected from the sidebar (Figure 1). The active page is highlighted; the remaining two are one click away. Each page is self-contained: inputs on that page do not carry over to the others.

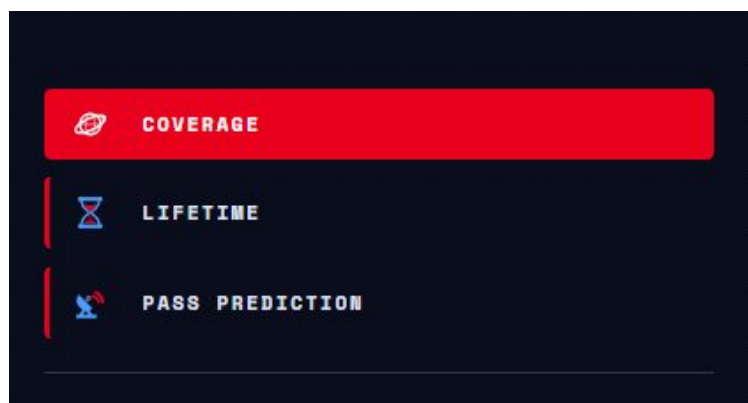


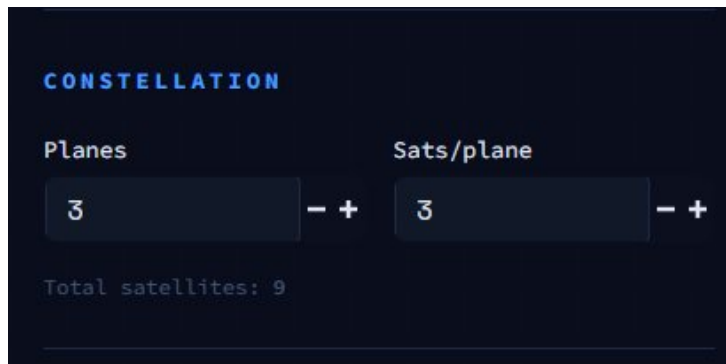
Figure 1: Sidebar navigation. The three modules — Coverage, Lifetime, and Pass Prediction — are independent pages. The active page (here, Coverage) is highlighted.

A.1 Coverage Page

The Coverage page computes access and revisit statistics for a constellation over either a Region of Interest or a set of ground stations. Inputs are entered top-to-bottom in the sidebar; the analysis runs only when **Run Analysis** is pressed at the bottom.

A.1.1 Constellation

The constellation is defined by the number of orbital **Planes** and the number of **Satellites per plane** (Figure 2). The total satellite count is derived as $N = P \times S$ and shown live beneath the inputs. Satellites are spread evenly in mean anomaly within each plane, and planes are spread evenly in RAAN, following the Walker-style spacing of Equations (3)–(4). In the worked example, 3 planes $\times$ 3 satellites gives $N = 9$.



The screenshot shows a dark-themed interface with the title 'CONSTELLATION' in blue. Below the title, there are two input fields: 'Planes' and 'Sats/plane'. Both fields have a value of '3' entered. To the right of each input field are minus and plus buttons. Below these fields, the text 'Total satellites: 9' is displayed in a light blue color.

Figure 2: Constellation input. Planes and satellites-per-plane define the Walker geometry; the total ($N = P \times S = 9$) updates live.

A.1.2 Orbital Elements

This block sets the shared orbit for all satellites (Figure 3). Entering the **Altitude** immediately derives and displays the semi-major axis, orbital period, and nodal-regression (J_2 drift) rate — here 500 km gives an SMA of 6878.1 km and a period of 94.62 min.

With **Auto SSO inclination** enabled, the inclination required for a Sun-synchronous orbit at that altitude is computed from Equation (7) and shown read-only (97.4016° at 500 km). Disabling the toggle lets the inclination be entered manually.

The **Epoch date and time** fix the reference instant for the Sun’s position. The **LTAN** (Local Time of Ascending Node) and **RAAN₀** (RAAN of the first plane) fields are *live-linked* via Equations (5)–(6): editing one updates the other for the given epoch, so the orbit can be specified in whichever convention is more natural. In the example, an LTAN of 10.50 h corresponds to a RAAN₀ of 320.68°.

ORBITAL ELEMENTS

Altitude (km)

500 - +

SMA 6878.1 km · Period 94.62 min · J2 drift +0.9856 °/day · e = 0.001

☒ Auto SSO inclination

Inclination: 97.4016° (J2-derived SSO)

Epoch date (UTC)	Epoch time (UTC)
2026/03/02	08:23

LTAN (decimal hours) ③

10.50 - +

RAAN₀ (°) ③

320.6800 - +

Figure 3: Orbital elements. Altitude derives SMA, period, and J_2 drift; Auto SSO inclination applies Equation (7); LTAN and RAAN₀ are live-linked through the epoch.

A.1.3 Simulation Window

The simulation block (Figure 4) sets the time span and sampling. Two repeat-ground-track figures are displayed for reference: the **exact repeat cycle** (here 155 days) and the **approximated repeat cycle** within the 30-day search window (9 days; see Section 3.3). Ticking **Use repeat cycle as simulation duration** sets the duration to that approximated cycle automatically; otherwise the **Duration** is entered directly.

The **Sample step** trades accuracy against runtime — 30 s (fine), 60 s (standard), or 120 s (fast). The resulting stop time and total step count are shown beneath (a 7-day run at 60 s yields 10,081 steps). Smaller steps reduce the time-step quantisation error of Section 3.3’s companion table.

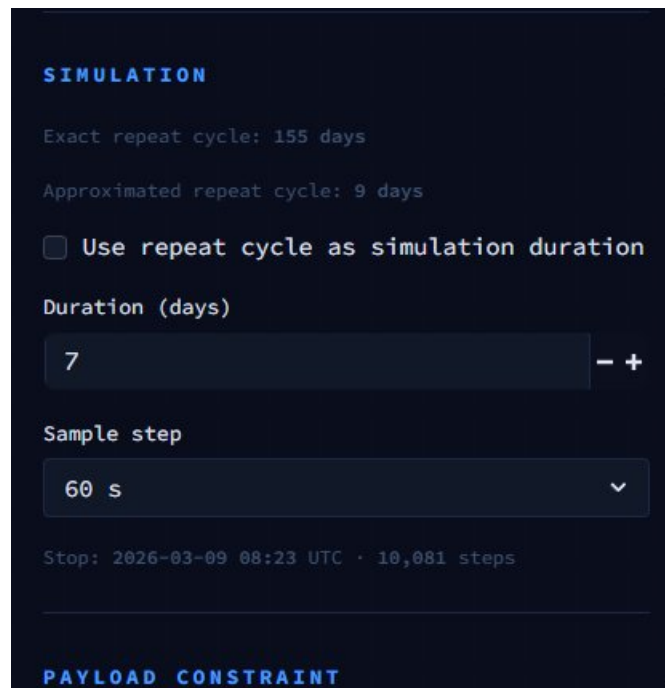


Figure 4: Simulation window. The exact and approximated repeat cycles are shown for reference; duration and sample step set the span and resolution, with the derived stop time and step count displayed.

A.1.4 Payload Constraint

The **Max off-nadir angle** slider (Figure 5) sets the payload’s pointing limit for Region-of-Interest access — this is an Earth-observation viewing-geometry constraint, *not* a communications link constraint. A ground point counts as observed only when the satellite is within this off-nadir angle of it, equivalent to a minimum elevation of $90 - \theta_{\max}$ as given in Equation (8). At $\theta_{\max} = 45$, the equivalent minimum elevation for the ROI is 45° . (Ground-station access, by contrast, uses each station’s own elevation mask rather than this slider.)



Figure 5: Payload constraint. The maximum off-nadir angle defines ROI viewing geometry; the equivalent minimum elevation ($90 - \theta_{\max}$) is shown live. This applies to ROI targets only, not ground stations.

A.1.5 Ground Stations

Ground stations are managed as an editable list (Figure 6). Two defaults are pre-loaded — Helmos Observatory (NOA) and the UoA ground station at Psachna — and either can be expanded to

edit its name, latitude, longitude, altitude, and minimum elevation, or removed entirely. **Add ground station** appends a custom station. Each station is evaluated against its own elevation mask using the geometry of Equation (9).



Figure 6: Ground stations. Pre-loaded defaults can be edited or deleted, and custom stations added. Each carries its own minimum-elevation mask.

A.1.6 Region of Interest and Running the Analysis

The Region of Interest defines the area over which coverage is gridded (Figure 7). Two sources are available:

- **Built-in country outline** — select from the supported countries (all EU states plus selected Mediterranean countries), whose borders are hardcoded from Natural Earth 1:110m polygons. Fast and offline, but coarse near coastlines.
- **Custom boundary shapefile** — upload a `.shp` (with companion `.dbf/.shx`) for higher-resolution borders. Per-country shapefiles are freely available online and give noticeably better coastline fidelity than the built-in polygons.

The **Rows** and **Cols** fields set the density of the sampling grid over the region’s bounding box; the count of interior grid points (those actually inside the border, via the ray-casting test of Equation (8)’s companion algorithm) is shown live — here a 5×10 grid yields 13 interior points over Greece. The **Single point** toggle collapses the ROI to one coordinate, useful for validating a specific target.

Pressing **Run Analysis** executes the propagation and returns the coverage and revisit metrics described in Section 3.

REGION OF INTEREST

Upload custom boundary shapefile ?

Upload 200MB per file • SHP, DBF, S...

☐ Single point ?

Country (built-in outline) ?

Greece ▼

Rows: 5 − + Cols: 10 − +

Interior grid points: 13 (Greece)

▶ Run Analysis

Figure 7: Region of Interest. A built-in country outline or an uploaded shapefile defines the border; Rows and Cols set the grid density, with the live interior-point count shown. Run Analysis launches the computation.

A.1.7 Outputs

After **Run Analysis**, results are presented in two parts: ROI metrics and ground-station results.

The ROI view (Figure 8) opens with a **Propagation Summary** echoing the configuration that produced the run (satellite count, SMA, eccentricity, inclination, $RAAN_0$, LTAN, off-nadir limit, step count, and the epoch span) so a result can always be traced back to its inputs. Below it, **ROI Metrics** aggregate over the whole region: **Coverage** (the percentage of grid points seen at least once), **Mean Access Ratio**, and the **Mean** and **Max Revisit** times. The **ROI Per-Point Breakdown** table then lists, for each interior grid point, its coordinates, number of passes, mean and maximum revisit gaps, access ratio, and peak elevation — letting coverage be inspected spatially rather than only in aggregate. The full table is exported with **Download ROI metrics CSV**.

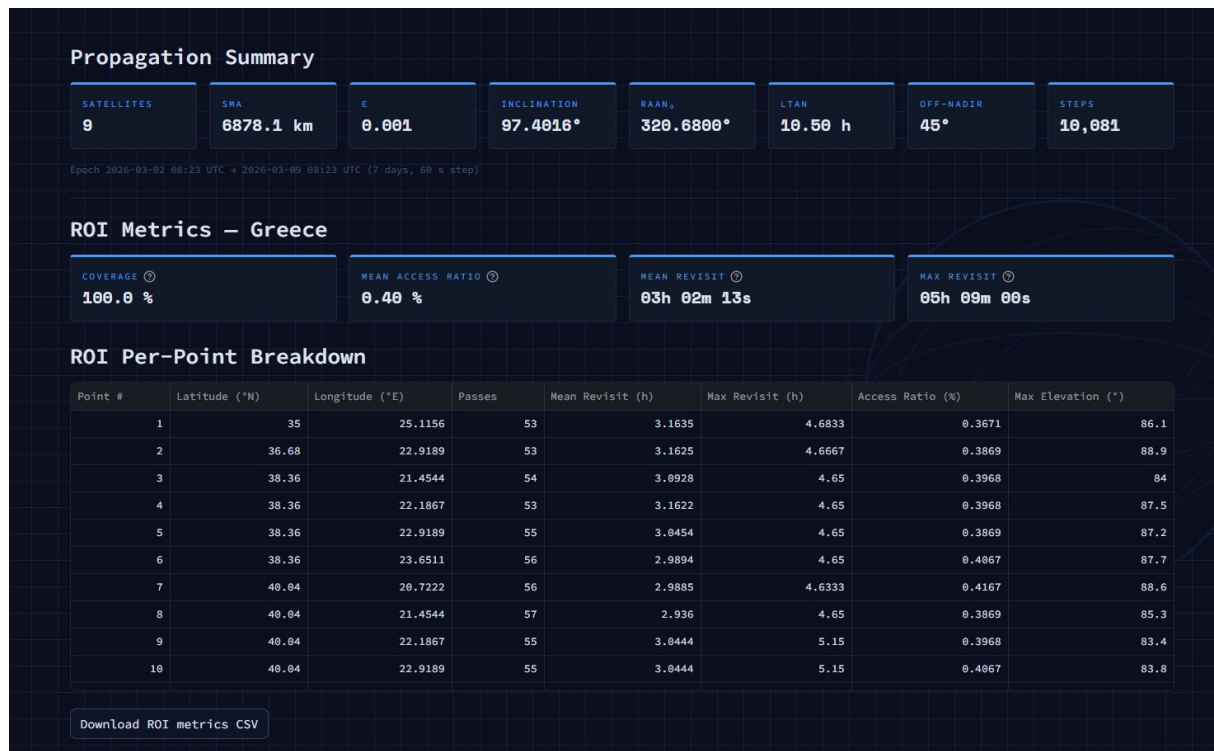


Figure 8: Coverage ROI output. A propagation summary traces the run back to its inputs; aggregate ROI metrics and a per-point breakdown table quantify coverage spatially. The table is downloadable as CSV.

The **Ground Station Results** view (Figure 9) presents one tab per station. Each carries its own summary — total **Passes**, **Access Ratio**, average pass **Duration**, and **Mean/Max Revisit** — above a pass list giving the start, end, duration, and peak elevation of every contact window. As with the ROI view, the complete pass list is exported via **Download pass list CSV**.

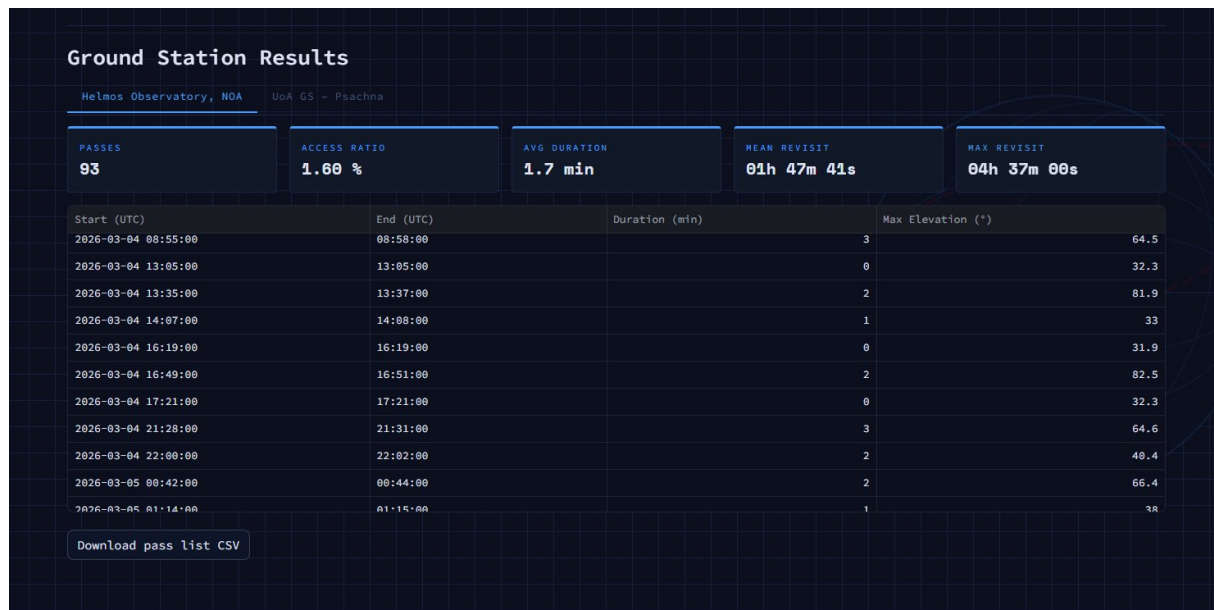


Figure 9: Coverage ground-station output. One tab per station, each with summary statistics and a full pass list (start, end, duration, peak elevation), downloadable as CSV.

A.2 Lifetime Page

The Lifetime page estimates how long a satellite remains in orbit before atmospheric drag brings it below the decay threshold, and optionally how a propulsive disposal manoeuvre shortens that time. Unlike the Coverage page, it performs no orbital propagation: it integrates the single scalar decay equation of Section 4.4. This has one important consequence for the inputs, noted below.

A.2.1 Satellite Parameters

The drag response of the satellite is set by three inputs (Figure 10): **Dry mass**, **Cross-sectional area** A , and **Drag coefficient** C_D (default 2.2). From these the tool derives the ballistic coefficient $\beta = m/(C_D A)$, which governs the decay rate — a higher β decays more slowly. The **Wet mass** (dry mass plus any propellant from the Propulsion block) is shown live; with propulsion disabled it equals the dry mass, as in the worked example (5.50 kg, 0.03 m², $C_D = 2.2$).

SATELLITE PARAMETERS

Dry mass (kg)
5.50 - +

Cross-sectional area (m²)
0.0300 - +

Drag coefficient Cd
2.20

Wet mass: 5.50 kg

Figure 10: Satellite parameters. Dry mass, cross-sectional area, and drag coefficient define the ballistic coefficient $\beta = m/(C_D A)$. Wet mass updates live as propellant is added.

A.2.2 Initial Orbit

The **Altitude** sets the initial circular orbit (Figure 11). At 500 km the page displays a warning that NRLMSISE-00 density is at the lower boundary of reliable accuracy and that results there should be treated as indicative, with DRAMA OSCAR recommended for high-altitude work — consistent with the validation gap reported in Section 4.7 and the ceiling noted in Limitation 8. The **Eccentricity** is fixed at 0.001 and shown as a stated assumption: the integrator treats the orbit as circular.

The orbit is considered decayed once it drops below a fixed **deorbit threshold** of 80 km (DRAMA OSCAR convention).

Why there is no inclination input. Inclination is deliberately *not* an input on this page. The estimator integrates a scalar da/dt (Section 4.4) and samples the atmosphere at the equator ($\phi = 0$); inclination never enters that calculation. Exposing an inclination control would imply a dependence that does not exist, so it is omitted — the page models drag-driven altitude decay, not orbital geometry.

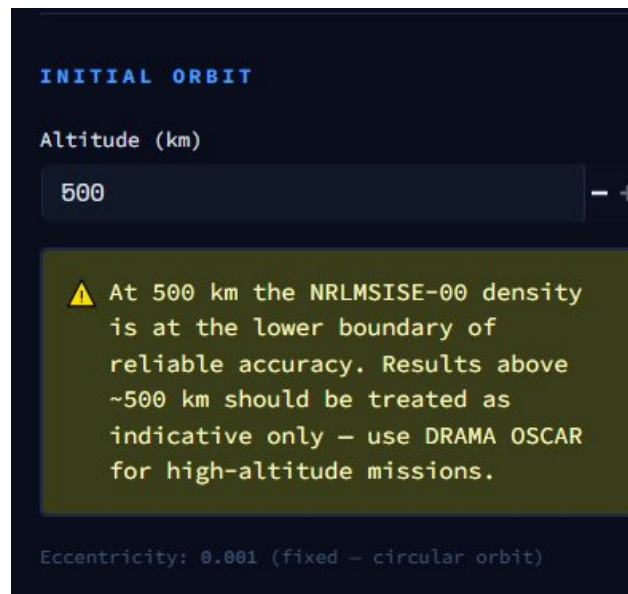


Figure 11: Initial orbit. Altitude sets the starting circular orbit, with a boundary-accuracy warning at 500 km; eccentricity is a fixed assumption, and the deorbit threshold is fixed at 80 km.

A.2.3 Solar Activity

Atmospheric density depends strongly on solar activity, selected here from the ECSS-E-ST-10-04C scenarios (Figure 12): **Solar Min**, **Mean**, **Max**, or **All three** (which runs and overlays all scenarios for a bounding range). The corresponding F10.7, F10.7a, and Ap indices are shown live for the chosen scenario — Solar Mean gives $F10.7 = F10.7a = 140$ and $Ap = 15$, matching the values tabulated in Section 4.3. Higher solar activity inflates thermospheric density and shortens lifetime, so running all three scenarios brackets the uncertainty from future solar conditions.

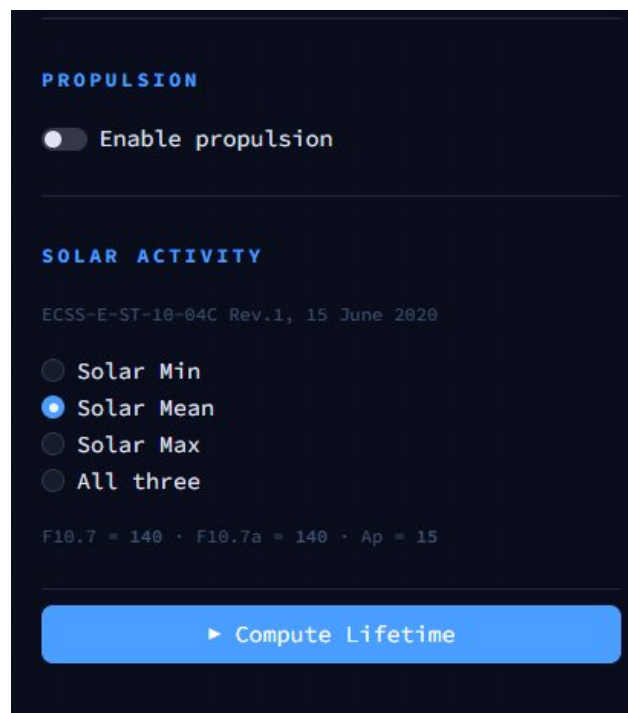


Figure 12: Solar activity. The ECSS Min/Mean/Max scenarios (or all three) set the F10.7 and Ap indices that drive atmospheric density. Solar Mean is shown, with its indices displayed live.

A.2.4 Propulsion and Running the Estimate

Propulsion is optional. With **Enable propulsion** off, the page reports the natural-decay lifetime. Enabling it (Figure 13) exposes the **Propellant mass**, **Thrust T** , and **Specific impulse I_{sp}** , and a **Deorbit strategy** selector with two options:

- **Two-burn circularization** — a Hohmann-style disposal to a lower circular **Target altitude**, after which the orbit decays naturally (Section 4.5.1). The two impulses and total ΔV are shown live; in the example, a $500 \rightarrow 450$ km transfer gives $\Delta V_1 = \Delta V_2 = 13.9$ m/s for a total of 27.8 m/s.
- **Continuous thrust** — a low-thrust spiral, integrated with the modified decay equation of Section 4.5.2.

The propellant required for the selected strategy is computed via Tsiolkovsky and checked against the available propellant mass. Pressing **Compute Lifetime** runs the integration and returns the decay time, the 5-year compliance status (Section 4.6), and the altitude-versus-time profile.

PROPULSION

☒ Enable propulsion

Propellant mass (kg)

0.500 - +

Thrust T (mN)

1.00 - +

Specific impulse I_{sp} (s)

200 - +

Deorbit strategy

Two-burn circularization ▼

Target altitude (km)

450 - +

ΔV burn 1 (perigee drop): 13.9 m/s

ΔV burn 2 (circularization): 13.9 m/s

Total ΔV : 27.8 m/s

Figure 13: Propulsion. When enabled, propellant, thrust, and I_{sp} are entered and a deorbit strategy chosen. Two-burn circularization to a target altitude is shown, with the live ΔV breakdown (here $13.9 + 13.9 = 27.8$ m/s for $500 \rightarrow 450$ km).

A.2.5 Outputs

Pressing **Compute Lifetime** produces a decay plot and a set of result cards. With the *All three* solar scenario selected, the plot (Figure 14) overlays the altitude-versus-time decay for Solar Min, Mean, and Max on one set of axes, each annotated with its lifetime in days. Two reference lines are drawn: the 200 km step-size boundary (below which the integrator switches to a finer step, Section 4.4) and the selected deorbit threshold (80 km here). The strong spread between the

curves — in the example, from 576 days at Solar Max to 32,700 days at Solar Min — illustrates directly how sensitive lifetime is to solar activity, which is the motivation for running all three scenarios.

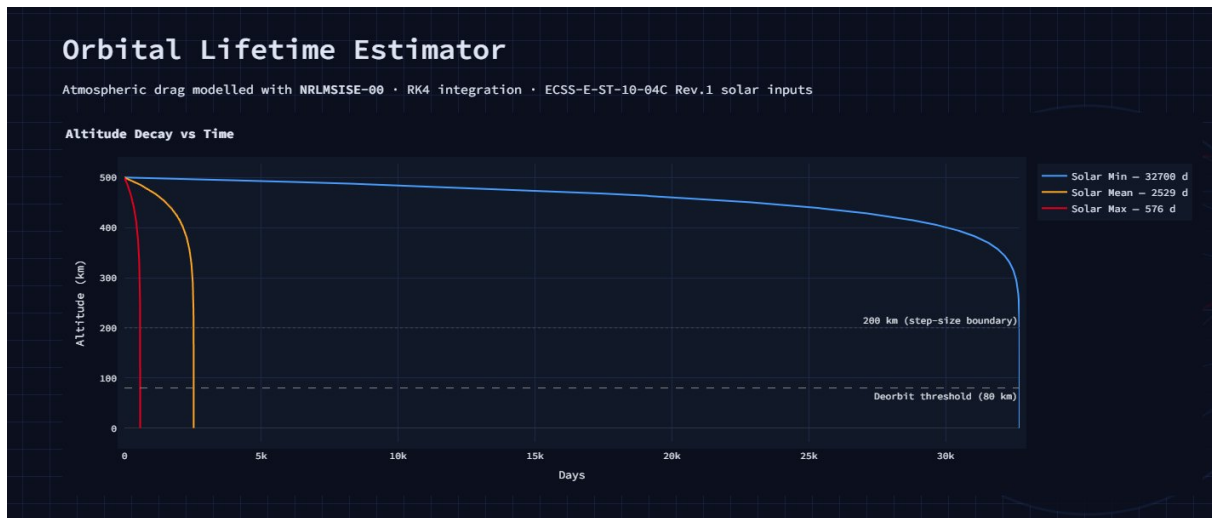


Figure 14: Lifetime decay plot. Altitude versus time for the three solar scenarios overlaid, with the 200 km step-size boundary and the deorbit threshold marked. The order-of-magnitude spread between Solar Max and Min shows the dominant role of solar activity.

Below the plot, one **Results** card per scenario (Figure 15) reports the lifetime in both days and years, the calendar **Deorbit date**, and the **Status**. Each card carries the 5-year compliance verdict of Section 4.6 as a coloured banner — in the example, Solar Min and Mean are flagged non-compliant (89.53 and 6.92 years), while Solar Max is compliant (1.58 years). The footnote cites the governing requirements (ESA ESSB-ST-U-007, the FCC Report and Order, and the IADC guideline) together with the atmosphere, drag, and solar-model references, so the verdict is traceable to its standard.



Figure 15: Lifetime result cards. Per-scenario lifetime (days and years), deorbit date, and status, each with a 5-year compliance banner. The footnote cites the governing debris-mitigation requirements.

A.3 Pass Prediction Page

The Pass Prediction page computes ground-station contact windows for an existing satellite defined by a TLE. It reuses the same elevation and window-detection geometry as the Coverage page (Section 5), but is driven by a real TLE rather than idealised Keplerian elements, and runs for a single satellite at a fine 10-second step.

A.3.1 TLE Input

The satellite is supplied as a Two-Line Element set, uploaded as a `.txt` file in standard 2-line or 3-line format (Figure 16). The TLE epoch embedded in the file becomes the start of the prediction window (see Time Window below), since that is the instant at which the elements are most accurate.

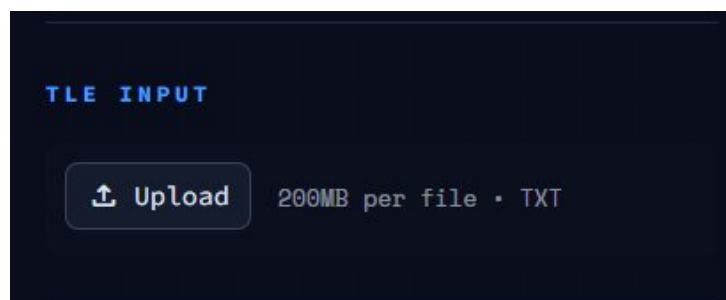


Figure 16: TLE input. A standard 2- or 3-line element set is uploaded as a `.txt` file; its epoch anchors the prediction window.

A.3.2 Ground Location

The observer is set either as a **Preset station** — the same Helmos and Psachna defaults as the Coverage page, with their coordinates shown on selection — or as **Custom coordinates** entered by hand (Figure 17). The **Min elevation** field sets the horizon mask: a pass is only reported once the satellite rises above this angle at the chosen location. In the worked example the observer is Helmos Observatory (37.9856°N, 22.1983°E) with a 5° mask.

GROUND LOCATION

☒ Preset station ☐ Custom coordinates

Station

Helmos Observatory, NOA

Lat 37.9856°N · Lon 22.1983°E

Min elevation (°)

5

► Predict Passes

Figure 17: Ground location. A preset station or custom coordinates define the observer; the minimum-elevation mask sets the horizon threshold for reporting passes.

A.3.3 Time Window and Running the Prediction

Once a TLE is loaded, the prediction window is shown (Figure 18). The simulation *starts at the TLE epoch* (here 2026-06-01 16:53 UTC), and **Predict passes for next** selects how far ahead to propagate — 1, 2, 3, 7, or 14 days. The resulting stop time and total step count are displayed (a 3-day window at the 10-second step gives 25,921 steps).

A **TLE age warning** reports the position and timing error expected for the chosen propagation span, following the accuracy bands of Section 5.3: a 10-day-old TLE, for example, is flagged with a ~5–20 km position error and a timing error under ~5 min. The note cites that these are derived from LEO orbital velocity and that real accuracy depends on the satellite’s altitude and drag environment.

Pressing **Predict Passes** runs the propagation and returns the contact windows — rise/set times, peak elevation, and duration — for the selected location and window.



Figure 18: Time window. The prediction starts at the TLE epoch; the days-ahead selector sets the span, with the derived stop time, step count, and a TLE-age accuracy warning shown. Predict Passes launches the computation.

A.3.4 Outputs

Predict Passes returns a summary, a pass schedule, and an elevation profile. The summary header (Figure 19) reports the **Total passes** in the window, the **Next pass** time, its **Next max elevation**, and the **Mean duration**.

The **Pass Schedule** table lists every contact window, one row per pass, with the following columns:

- **Start / End (UTC)** — rise and set times, defined as the instants the satellite crosses the minimum-elevation mask.
- **Duration (min)** — length of the contact window.
- **Max El (°)** — peak elevation reached during the pass; the higher this is, the closer the satellite comes to passing directly overhead.
- **Az at Max El (°)** — azimuth at the moment of peak elevation, indicating the direction to point at culmination.
- **Quality** — a convenience rating derived from peak elevation (e.g. *Overhead*, *Good*, *Low*, *Very Low*), flagging at a glance which passes are usable for a given link.

The complete schedule is exported with **Download passes as CSV**.



Figure 19: Pass Prediction schedule output. Summary cards report total passes and the next pass; the table gives rise/set times, duration, peak elevation, azimuth at culmination, and a quality rating per pass. Downloadable as CSV.

The **Elevation Profile** (Figure 20) plots elevation against time for the whole window, with each pass drawn as a separate trace and the minimum-elevation mask marked as a horizontal reference. The characteristic narrow spikes show how briefly a LEO satellite is above the horizon on each pass, and their heights give an immediate visual comparison of pass geometry across the window. The figure footnote cites the propagation and TLE-accuracy references (Sections 2 and 5.3).



Figure 20: Pass Prediction elevation profile. Elevation versus time across the window, one trace per pass, with the minimum-elevation mask marked. Spike heights compare pass geometry at a glance.